Lecture 16:The Financial System, Money and Interest Rates

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"The value of money is not in the coin or the note, but in the ability to create value."

Lecture overview

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- The Meaning and Functions of Money
- The Financial System
- The Supply of Money
- The Demand for Money
- Equilibrium
- Money, Aggregate Demand and Inflation

The Meaning and Functions of Money

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The Meaning and Functions of Money

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- Money includes more than cash—mainly bank deposits.
- Most transactions are electronic (cards, cheques, apps), not in cash.
- The functions of money
  - **Medium of exchange:** Simplifies trade vs. barter system.
  - **Store of wealth**: Allows savings for future use.
  - **Unit of account**: Enables price comparisons and adds up value (e.g. GDP).
  - **Standard of deferred payment:** Agrees future payments in monetary terms.What should count as money?
- What should count as money?
  - narrow definitions of money
  - broad definitions of money

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The Financial System

The Financial System and Financial Intermediation

- Financial institutions act as intermediaries between savers and borrowers.
- Key roles:
  - **Expert advice**: Guides clients on investments and financing.
  - **Channeling funds**: Directs money to high-return areas, boosting efficiency.
  - **Maturity transformation:** Converts short-term deposits into long-term loans.
  - **Risk transformation:** Diversifies and manages lending risk.
  - **Payment transmission:** Facilitates cashless payments (cards, direct debits, etc.).

Profitability, Liquidity and Capital Adequacy

- **Profitability**: Banks profit by lending at higher rates than they pay on deposits.
- **Liquidity**: Ability to convert assets to cash easily; cash is most liquid.
- Banks balance between:
  - Profitable but illiquid assets (e.g. loans, mortgages).
  - Liquid but less profitable assets (e.g. cash, short-term loans).

Profitability, Liquidity and Capital Adequacy

- **Liquidity ratio**: Proportion of liquid assets to total assets; must be optimized to avoid crisis or low profit.

- **2008 crisis**: Banks increased liquidity ratios due to fears of bad debt.

- **Basel III**: Introduced **Liquidity Coverage Ratio (LCR)** to ensure banks hold enough high-quality liquid assets (HQLAs) to cover 30-day cash outflows, increasing from 60% to 100% by 2022.

Q The liquidity ratio of a bank refers to its ratio of:

A. Treasury bills to government bonds.

B. liquid to illiquid assets.

C. cash to advances.

D. liquid assets to total liabilities.

E. total assets to total liabilities.

The Central Bank

- Every country has a central bank (e.g., Bank of England, ECB, Fed).
- Two main roles:
  - Oversee the monetary system and maintain stability.
  - Act as the government’s banker and implement monetary policy.
- Bank of England gained operational independence in 1997, now sets interest rates.
- ECB and Fed are also independent central banks.

The Central Bank

Key functions of the Bank of England:

- **Issues banknotes** (sole issuer in England and Wales).
- **Acts as a banker:**
  - To the government (manages Exchequer and National Loans Fund).
  - To banks (holds reserve balances and helps manage liquidity).
  - To overseas central banks (holds sterling deposits).
- Operates monetary policy via:
  - Interest rate setting (by the MPC).
  - Open-market operations (buying/selling securities to manage liquidity).
  - Quantitative easing (QE) to inject money into the financial system.

The Central Bank

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- Provides liquidity to banks through:
  - Index long-term repos (ILTRs).
  - Discount window facility (DWF).
  - Contingent term repo facility (CTRF) for emergencies.
- Oversees financial institutions: After 2013, regulation split into:
  - **FPC** (macro-prudential regulation),
  - **PRA** (prudential regulation of firms),
  - **FCA** (consumer protection & market regulation).

Manages exchange rate policy via foreign exchange reserves

The Supply of Money

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The Supply of Money

- Money supply must be measured to be monitored/controlled.
- Monetary base (narrow money) = notes + coins in circulation.
- Broad money includes:
    - Time and sight deposits,
    - Retail and wholesale deposits,
    - Bank and building society deposits.
- In the UK, broad money = M4; in Europe/USA, it's called M3.
- **Credit creation**: Banks can expand the money supply by increasing deposits through lending.

Credit Creation

- Banks hold:
  - Deposits as liabilities,
  - Balances with central bank (for liquidity) and advances to customers (to earn profit) as assets.

Credit Creation

- Operate with a liquidity ratio (e.g., 10%).
  - If total deposits = £100 billion:
    - £10 billion in central bank (liquidity),
    - £90 billion lent out.

Credit creation:
Banks' original balance sheet

Liabilities	£bn	Assets	£bn
Deposits (old)	100	Balances with central bank (old)	10
		Advances (old)	90
Total	100	Total	100

Government Spending Increases Bank Liquidity

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- Government spends £10 billion (e.g., on infrastructure).
- Payments deposited in banks → banks send cheques to central bank → central bank increases banks' balances by £10 billion.

Credit creation:
Initial effect of a new deposit of £10bn

Liabilities	£bn	Assets	£bn
Deposits (old)	100	Balances with central bank (old)	10
Deposits (new1)	10	Balances with central bank (new1)	10
		Advances	90
Total	110	Total	110

Banks Have Surplus Liquidity

- New total deposits = £110 billion; balances with central bank = £20 billion.
- New liquidity ratio = 18.2%.
- To maintain 10% ratio, only £11 billion is needed → £9 billion is now excess and can be lent out.

Credit creation:
Initial effect of a new deposit of £10bn

Liabilities	£bn	Assets	£bn
Deposits (old)	100	Balances with central bank (old)	10
Deposits (new1)	10	Balances with central bank (new1)	1
		Advances (old)	90
	___	Advances (new1)	_9_
Total	110	Total	110

The Credit Cycle Continues

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- Customers borrow and spend the £9 billion.
- Shopkeepers deposit the money; interbank clearing adjusts central bank balances.
- The cycle of lending continues:
    - 10% (e.g., £0.9 billion) held back,
    - 90% (e.g., £8.1 billion) can be re-lent.
- This multiplies money through successive rounds of deposits and lending.

Credit creation:
Initial effect of a new deposit of £10bn

Liabilities	£bn	Assets	£bn
Deposits (old)	100	Balances with central bank (old)	10
Deposits (new1)	10	Balances with central bank (new1)	1
Deposits (new2)	9	Balances with central bank (new2)	0.9
		Advances (old)	90
		Advances (new1)	9
		Advances (new2)	8.1
Total	119	Total	119

•Process repeats, multiplying credit and deposits.
•An initial £10bn deposit can lead to a total £100bn increase in money supply.
•Known as the **bank (or bank deposits) multiplier** effect.

The Bank Deposits Multiplier

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- When cheques clear, surplus balances remain in the central bank.
- 10% liquidity is retained; 90% is available for lending.
- Process repeats, multiplying credit and deposits.
- An initial £10bn deposit can lead to a total £100bn increase in money supply.
- Known as the bank (or bank deposits) multiplier effect.

Credit creation:
Full effect of a new deposit of £10bn

Liabilities	£bn	Assets	£bn
Deposits (old)	100	Balances with central bank (old)	10
Deposits (new: initial)	10	Balances with central bank (new)	10
(new: subsequent)	90	Advances (old)	90
		Advances (new)	90
Total	200	Total	200

The Bank Deposits Multiplier

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- £10bn initial deposit → £90bn new advances → £100bn total increase.
- With a 10% liquidity ratio, the multiplier is 10.
- Formula: Bank deposits multiplier = 1/L (e.g., 1/0.1 = 10).
- Illustrates cumulative causation—a small initial event causes a much larger final effect.
- Measures how many times greater the final deposits are compared to the initial.

Q If an extra £100m is deposited in the banking system and the bank multiplier is 5, by how much will credit expand?

A. £80m

B. £180m

C. £400m

D. £500m

Credit Creation in the Real World

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- Three complications affect credit creation:

1. Variable liquidity ratios:
   - Banks may raise or lower ratios based on risk or opportunity.
   - Example: 2008 crisis—banks hoarded cash due to sub-prime fears.
2. Demand-side factors:
   - Banks may offer loans, but customers may not want to borrow.
   - Lowering interest rates may deter depositors.
3. No fixed liquidity ratio:
   - Banks hold various liquid assets—makes the true liquidity ratio hard to define.

Credit Creation in the Real World

- Leakages from the banking system:
  - Public may withdraw cash, reducing deposit-based credit creation.
- Money multiplier vs. bank deposits multiplier:
  - Money multiplier = $\Delta Ms/\Delta Mb$.
  - Reflects real-world cash leakage and broader credit creation.

What Causes Money Supply to Rise?

- Five key causes:

1. Central bank actions:
   - E.g., quantitative easing increases the monetary base.
   - May not always translate into more lending.
2. Inflow of funds from abroad:
   - Increases bank deposits → boosts credit creation.
3. Public-sector deficit:
   - If funded by banks, it increases the money supply.
   - If funded by the public, no change in money supply.

What Causes Money Supply to Rise?

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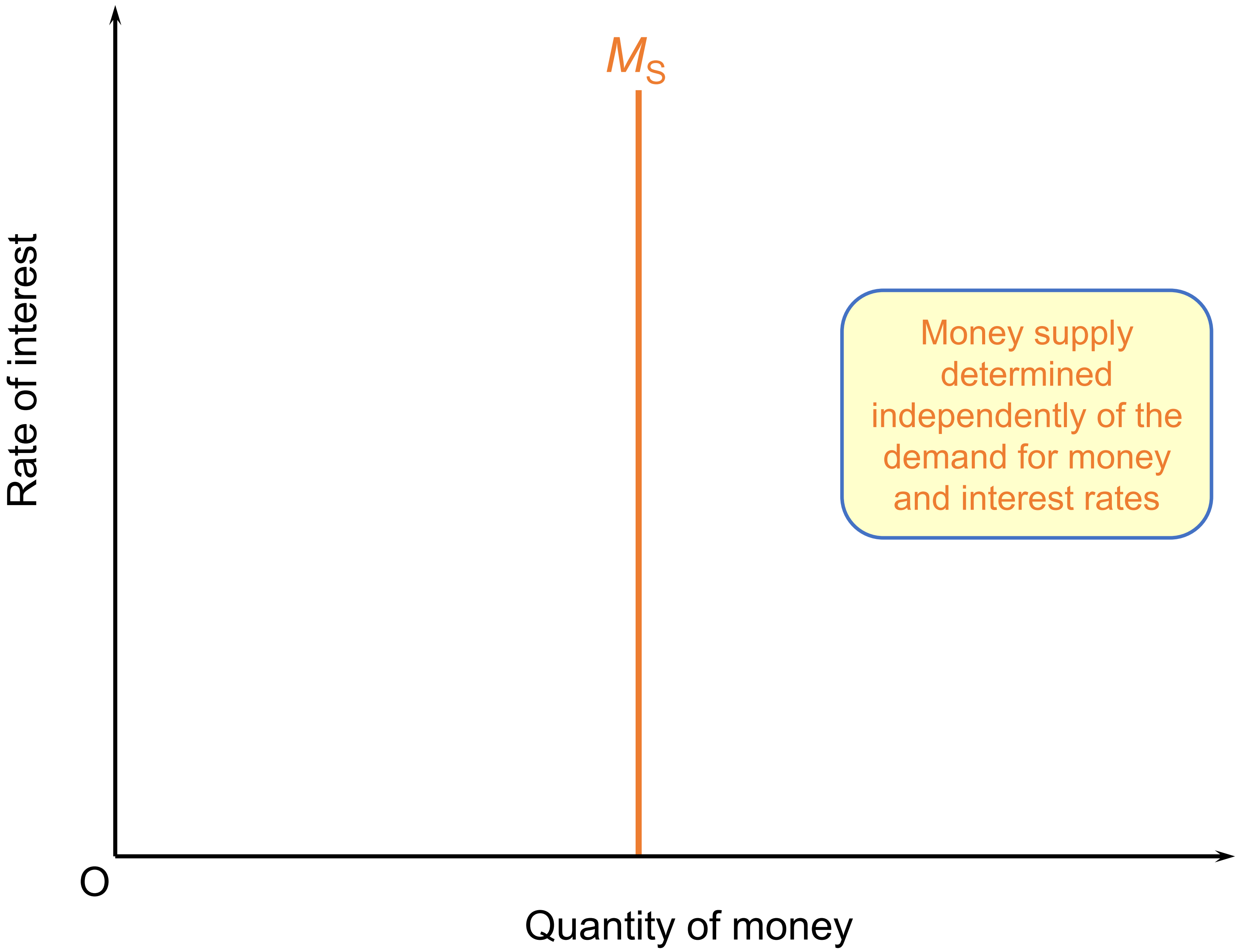
4. Lower bank liquidity ratios:
   - Increases surplus liquidity and loan creation.
   - Seen in trends like increased use of cards and interbank lending pre-2008.
5. Less cash held by public:
   - More deposits = more credit can be created by banks.

Money Supply & Interest Rates (Exogenous View)

- Classical theory: money supply is exogenous (set by central bank).
- Independent of interest rates.

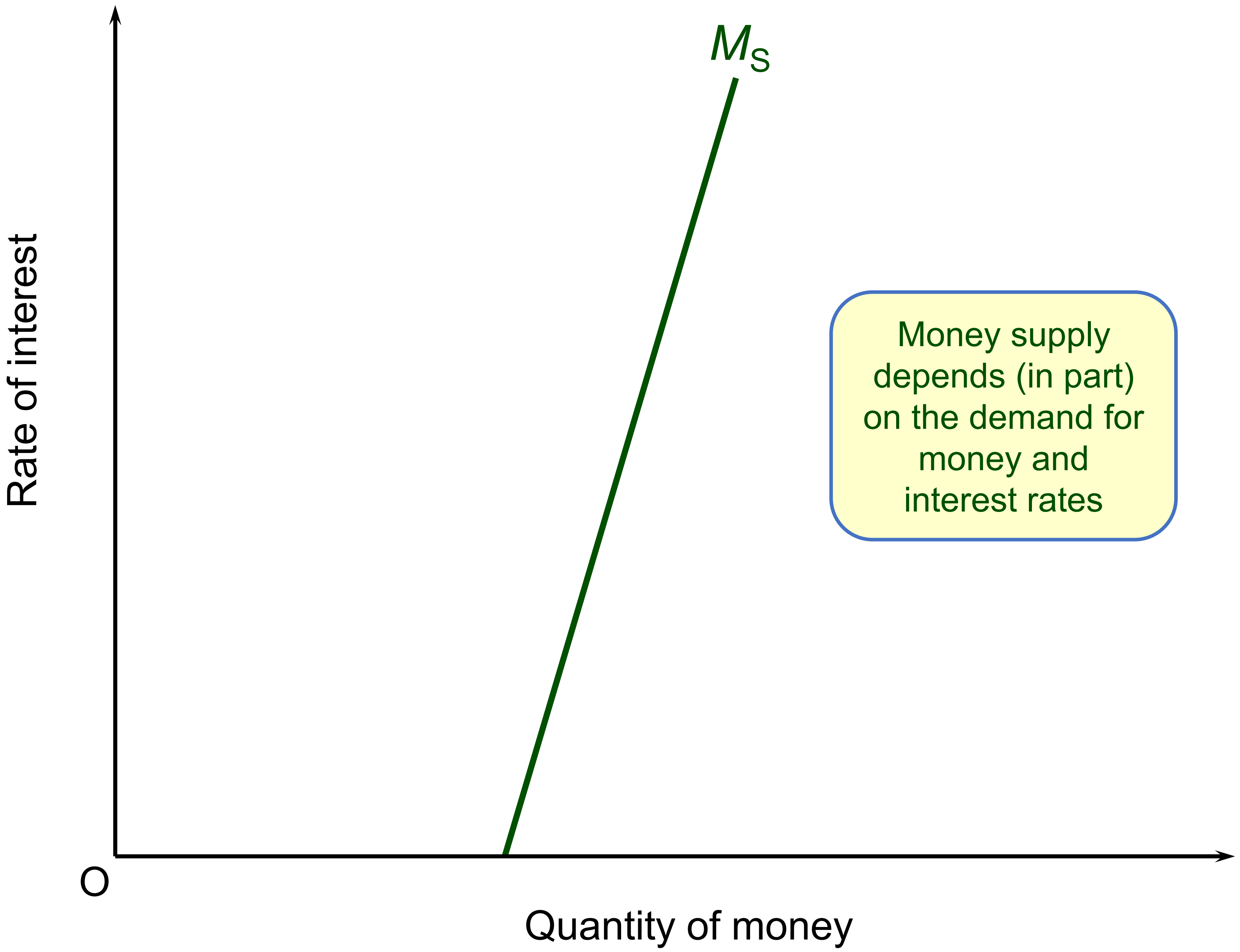
The supply of money curve: (a) exogenous money supply



Money Supply as Endogenous

- In reality, money supply is often endogenous (responds to demand).
- Higher interest rates may encourage banks to lend more.
- Banks may expand credit if they have liquidity or accept lower ratios.
- Some argue the money supply curve is horizontal—fully demand-driven.
- Bank confidence affects this relationship:
    - Optimistic → more credit.
    - Pessimistic → less credit, even if demand exists.

The supply of money curve: (b) endogenous money supply



Money supply depends (in part) on the demand for money and interest rates

Q Which of the following would NOT cause a rise in money supply (assume *ceteris paribus*)?

A. An increase in government spending financed by borrowing from the banking sector.

B. An increase in the proportion of the national debt financed by bills rather than by bonds.

C. The central bank imposes a statutory liquidity ratio on banks (above their current ratio).

D. The government finances the budget deficit by selling securities to the central bank.

The Demand for Money

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The Demand for Money

- Demand for money means holding wealth as money instead of spending or investing it.
- Three motives for holding money:
  - ***Transactions motive***: To conduct daily purchases and payments.
  - ***Precautionary motive***: For unexpected expenses or payment delays.
  - ***Speculative (asset) motive***: To avoid risk from volatile assets; money is safer but less rewarding.

Determinants of money demand:

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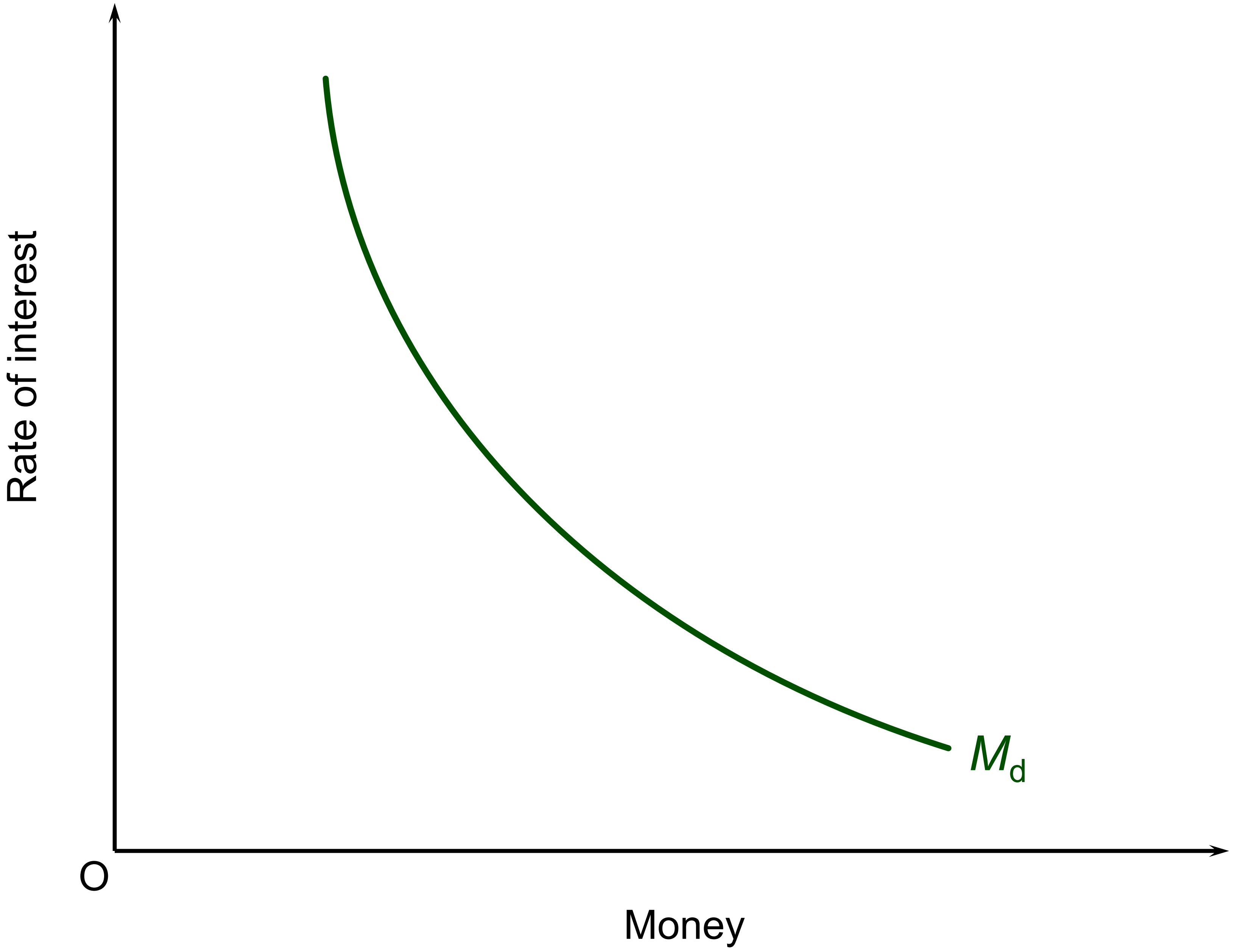
- **National income:** Higher income → more spending → higher money demand.
- **Payment frequency:** Less frequent pay → higher money balances needed.
- **Financial innovations**: Credit/debit cards, ATMs, and interest-bearing accounts influence money demand.
- **Expectations about future returns or exchange rates**: Fear of asset price falls or exchange rate movements increases money demand.
- **Interest rate**: Higher interest → lower money demand (higher opportunity cost of holding money).

The Demand-for-Money Curve

- Inverse relationship between interest rate and money demand.
- Interest rate rise → shift from holding money to assets (lower demand for money).
- Demand-for-money curve slopes downward.
- Movements along the curve: Due to interest rate changes.
- Shifts of the curve: Due to changes in income, payment habits, innovations, expectations, etc.

The demand for money curve



Q What do we mean by the term the 'demand for money'? Is it:

A. the proportion of people's income held as wealth?

B. a means of controlling the money supply?

C. a term used by the Bank of England to refer to the demands placed upon it by the banking sector?

D. the desire by individuals and firms to spend, given their level of income?

E. the demand to hold assets in money form?

Equilibrium

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Equilibrium in the Money Market

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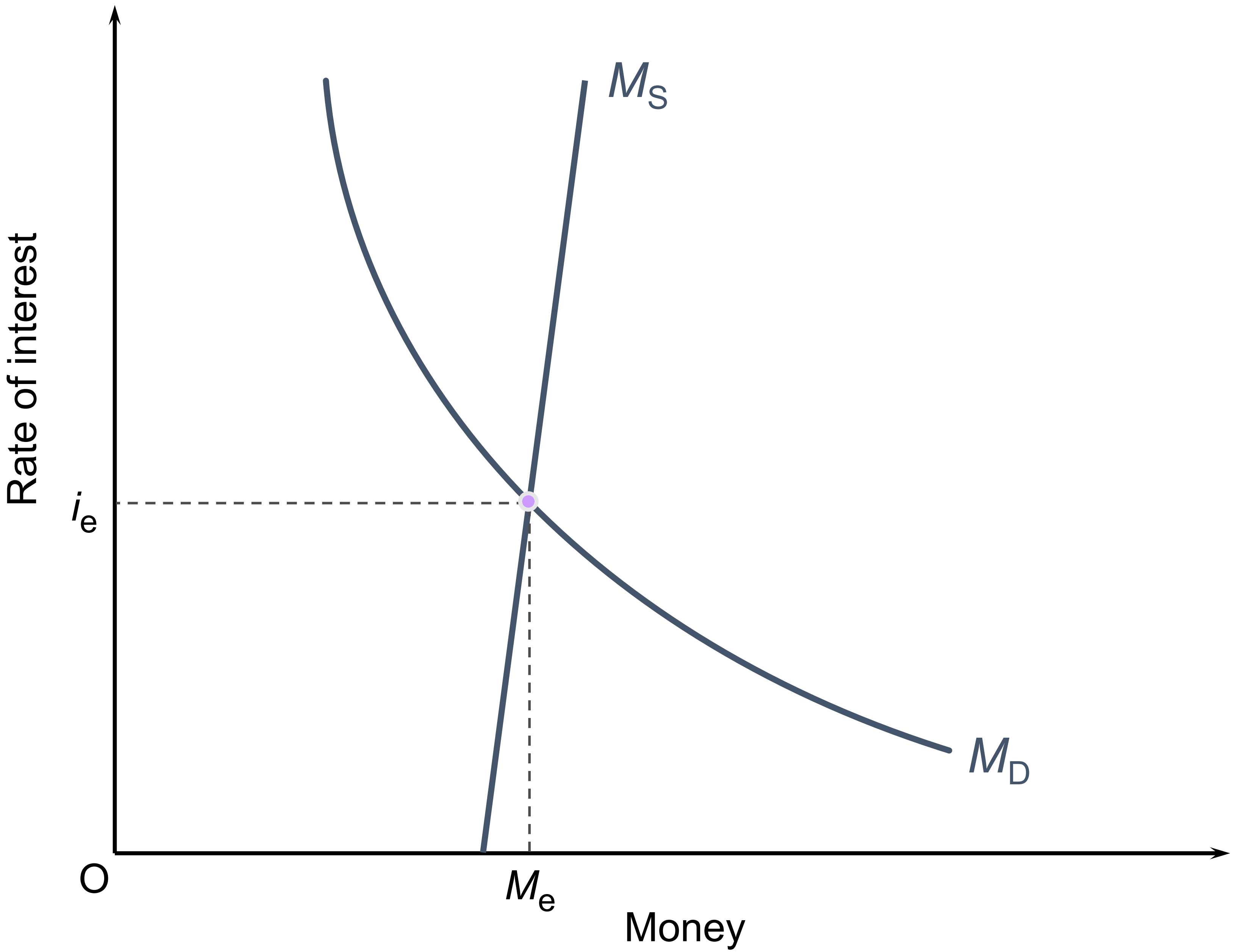
- Occurs when money demand (Md) = money supply (Ms)
- Equilibrium achieved through changes in the interest rate.

Equilibrium in the money market



Interest Rate Adjustment Process

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- If interest rate > equilibrium:
    - People buy assets → asset prices rise → interest rates fall.
- If interest rate < equilibrium:
    - People sell assets → asset prices fall → interest rates rise.
- The system adjusts until Md = Ms at a new equilibrium interest rate.

Shifts in Ms or Md

- Shift in Ms or Md leads to a new equilibrium interest rate and money quantity.
- Increase in Ms → interest rates fall.
- Increase in Md → interest rates rise.

Monetary Equilibrium

- Effects of changes in money supply on national income: the interest rate transmission mechanism
  - effect on interest rates of a rise in money supply
    - $Ms\uparrow \rightarrow r\downarrow$
  - effects of changes in interest rates on investment and saving
    - $r\downarrow \rightarrow I\uparrow, S\downarrow$
  - effects of changes in investment and saving on national income
    - $I\uparrow, S\downarrow \rightarrow Y\uparrow$
  - the size of these effects is keenly debated by economists

Monetary Equilibrium

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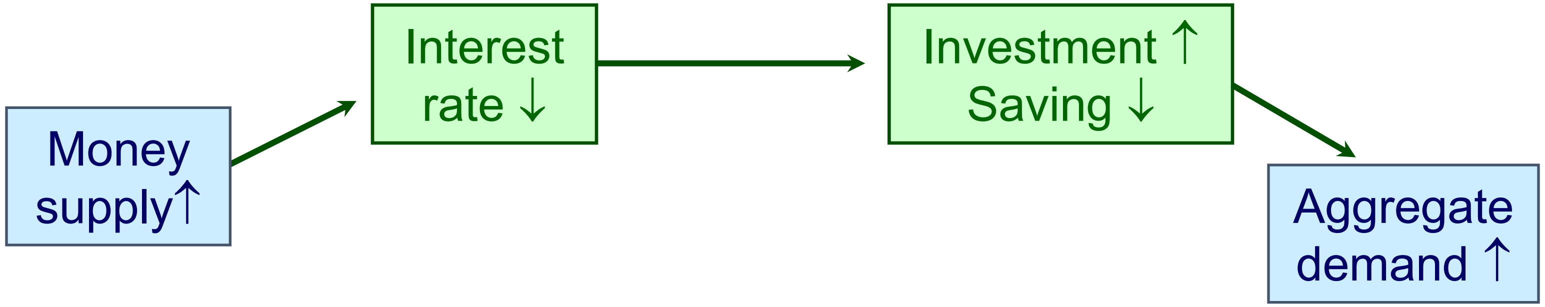
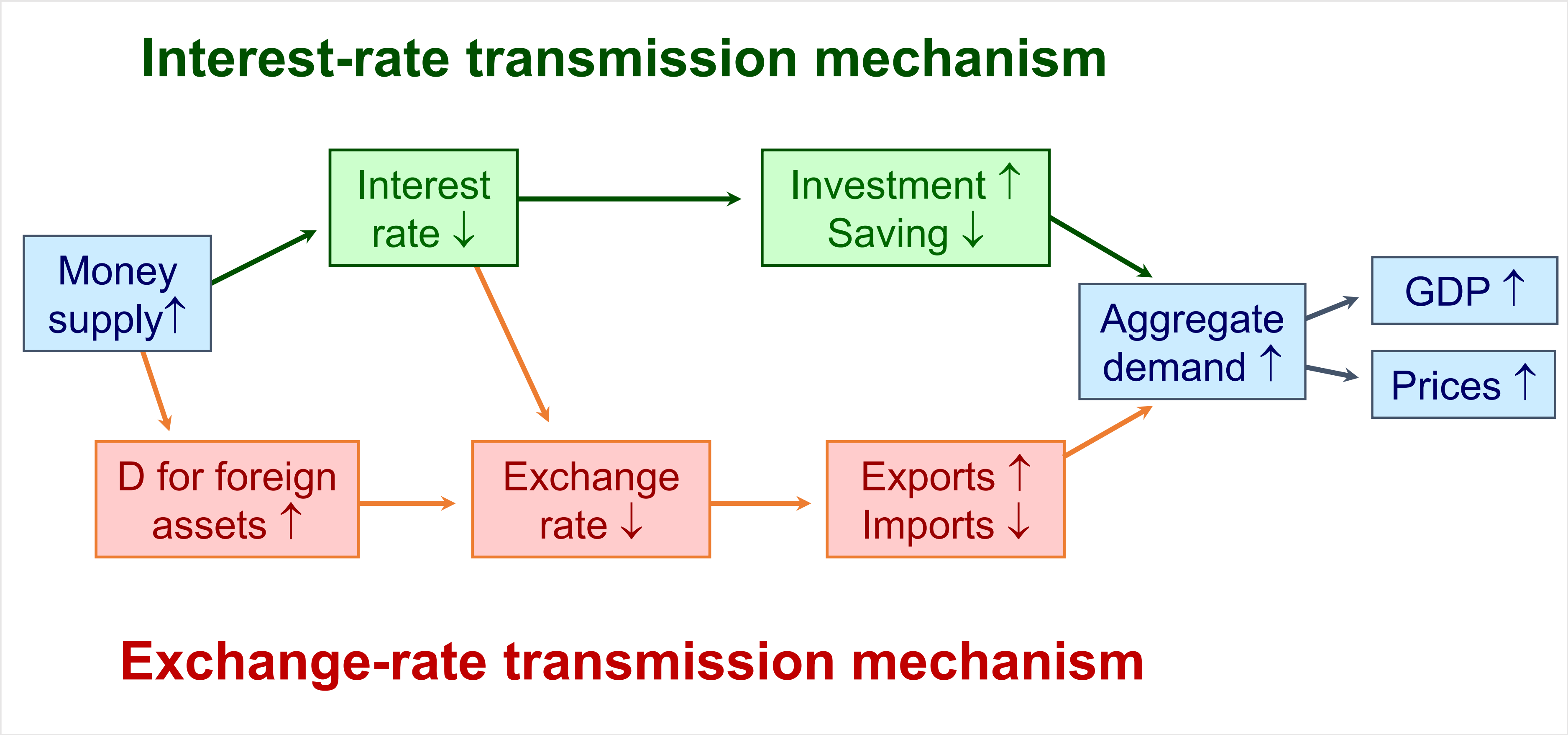
- Effects of changes in money supply on national income: the exchange rate transmission mechanism
  - effect on interest rates of a rise in money supply
    - $Ms\uparrow \rightarrow r\downarrow$
  - effects of changes in interest rates and a purchase of foreign assets on the exchange rate, and imports and exports
    - $r\downarrow \rightarrow er\downarrow \rightarrow X\uparrow, M\downarrow$
  - effects of changes in investment, imports and exports on national income
    - $I\uparrow, X\uparrow, M\downarrow \rightarrow Y\uparrow$

Monetary transmission mechanisms

Interest-rate transmission mechanism



Monetary transmission mechanisms



Q Assuming that interest rates and exchange rates are determined by the market, a reduction in money supply will result in:

A. a fall in both interest rates and exchange rates.

B. a rise in both interest rates and exchange rates.

C. a rise in interest rates and a fall in exchange rates.

D. a fall in interest rates and a rise in exchange rates.

Money, Aggregate Demand and Inflation

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Monetarist / New Classical View

- Strong belief in transmission mechanisms: money supply increase → higher aggregate demand → higher prices.
- Milton Friedman: Inflation is always caused by excessive money supply growth.
- In the long run, aggregate demand increases lead mainly to inflation, not higher output.

Keynesian View

- Looser relationship between money supply and inflation.
- Effect of money supply on output and prices depends on economic conditions (e.g., spare capacity, confidence).

Money Supply, Aggregate Demand and Inflation

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- Basic idea: More money → higher prices, assuming other factors constant.
- The equation of exchange: $MV = PY$
  - M money supply
  - V velocity of circulation
  - P price level (number of times greater than in base year)
  - Y real value of output at base-year prices
- $MV = PY$ means nominal spending = nominal GDP.

Money Supply, Aggregate Demand and Inflation

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Implications of $MV = PY$

- If M increases, then PY must also increase.
- Debate focuses on:
  - Whether increased M boosts Y (real output) or just P (prices).
  - Outcome depends on the aggregate supply response.

Q If M4 is £750bn, nominal GDP is £1500bn and the price level in base-year prices is 25,the velocity of circulation of M4 must be:

A. 2

B. 0.5

C. 60

D. 30

E. 6

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Questions